

Exam Date & Time: 19-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

Data Analytics with Python [NPT-3201]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) In a company, 60% of employees are male and 40% are female. 70% of the males and 50% of the females prefer work-from-home. Apply the concept of conditional probability to find the probability that a randomly chosen employee prefers work-from-home? (5)
 - A)
 - B)
- 2) A sample of 11 circuits from your large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms. Identify 95% confidence interval for the true mean resistance of the population. (5)
- 2) Explain the concept of Two-Way ANOVA and how it differs from One-Way ANOVA. Include the types of factors involved and a practical example where Two-Way ANOVA is applicable (5)
 - A)
 - B)
- 3) Build a two-sample t-test to compare the average salaries of employees in two departments of a company. Use a significance level of 0.05 and explain how you interpret the results. (5)
- 3) Prepare an analysis report on the concept of linear regression and its application in modelling the relationship between a dependent and independent variable. Also, list the assumptions made in linear regression analysis. (5)
 - A)
 - B)
- 4) Apply the concept of Maximum Likelihood Estimation to estimate the parameter of a normal distribution given a small sample of data. (5)
- 4) Analyse the role of the ROC curve in evaluating the performance of a binary classification model. What does the area under the ROC curve (AUC) indicate about the model's accuracy? (5)
 - A)
 - B)
- 5) Apply the Chi-Square Goodness of Fit Test to determine whether a die is fair based on the following observed frequencies. Use a 5% level of significance and interpret the results. (5)

Observed Frequencies:

Face of Die	1	2	3	4	5	6
Frequency	8	10	9	7	11	15

- 5) Distinguish between hierarchical clustering and K-Means clustering in terms of their approach to forming clusters, scalability, and ability to handle different types of data. (5)
- A)
- B) Define data analytics. What are the main types of data analytics? Give brief definitions any 2 (5)

-----End-----